

EC2 Auto Scaling with Load Balancer in VPC

Project Overview

This project sets up a scalable and highly available web server infrastructure on AWS using EC2 Auto Scaling, an Application Load Balancer (ALB), and a Virtual Private Cloud (VPC) with subnets.

Main Components:

- VPC with two public subnets across AZs
- EC2 Launch Template for web servers
- Auto Scaling Group (ASG) with scaling policies
- Application Load Balancer with listener and target group

Screenshots

Screenshots were previously included, but are currently missing after environment reset.

CloudFormation Template

```
AWSTemplateFormatVersion: '2010-09-09'  
Description: EC2 Auto Scaling with Load Balancer in a VPC  
...
```

EC2 User Data Script

```
#!/bin/bash  
yum update -y  
yum install -y httpd  
systemctl start httpd  
systemctl enable httpd  
echo "Hello from EC2 instance launched via Auto Scaling!" > /var/www/html/index.html
```